

Reservoir Water Level Prediction and Gate Operation Optimization Using Machine Learning: A Case Study of Shetrunji Reservoir, Gujarat

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Abstract

Control over reservoir water levels and gate settings at the spillways is key for preventing floods, protecting people living downstream, and efficient utilization of stored water especially when the reservoir is nearing its maximum capacity. This research project centers on the Shetrunji Reservoir in Gujarat, India, investigating the potential of machine learning to improve operations. Daily and event-based hydrological records for the reservoir covering the period 2020 to 2024 were compiled, digitized, and used to train and test predictive models. Precipitation behaviour, inflow characteristics, and the existing operating protocol were analysed, and three modelling frameworks were developed: computational automation of the current lookup-based practice; an Intermediate Machine Learning scheme that combines Quadratic Regression for the level-storage relationship with K-Nearest Neighbours regression for the gauge-discharge relationship; and an XGBoost regression model for direct water-level forecasting. On the held-out test partition, which comprises 20 percent of the observations,

the Quadratic Regression model reproduced the level-storage curve with a coefficient of determination of 0.9999, the K-Nearest Neighbours model reproduced the gauge-discharge curve with a coefficient of determination of 0.9995, and the XGBoost model forecast the reservoir water level with a root mean squared error of 0.0419 m, a coefficient of determination of 0.9995, and a mean squared error of 0.00175. These accuracies are interpreted in the light of the smooth, near-deterministic nature of the underlying rating relationships. A tiered gate-operation rule is proposed that extends the simple full-reservoir-level threshold by incorporating water-level bands, gate-sequencing priority, and structural safety considerations. The results indicate that data-driven models can support proactive flood-risk reduction and more consistent, timely gate operation, contributing to safer and more resilient reservoir management.

Keywords: Reservoir Operation, Machine Learning, Water Level Forecasting, Gate Control, Flood Mitigation, XGBoost.

1. Introduction

Among the most essential resources on Earth, water sustains the ecosystem, agriculture, and humanity itself. In fact, approximately 71 percent of the Earth's surface is covered by water in the form of oceans, rivers, lakes, glaciers, and aquifers, with water's capacity to exist in all three states being a factor behind the hydrological cycle which constantly renews it through processes of evaporation, condensation, and precipitation [1]. The fact that hydrological cycle connects the energy and carbon budgets of the planet makes it one of the major components of the global climate system [2]. The main way through which water is replenished is precipitation, which supports rivers, lakes, and aquifers, but it depends greatly on location and season. In the dry zones of India, rainfall is scant, unpredictable, and hardly forecastable [3], while long-term studies show decreasing tendencies of precipitation in the north and increasing in the south, with the hot and arid northwest region characterized by scarce rainfall and high evapotranspiration [4].

Rising demands are putting immense pressure on the supply of these resources. It is estimated that population growth, urbanization, and development will increase water requirements in cities, industries, and households greatly over the next few decades [5]. Currently, an estimated 2.3 billion people reside in nations experiencing water stress, while

approximately 4 billion people experience water scarcity in one month annually [6]. Freshwater resources are unevenly distributed due to various factors including geography, usage, pollution, and population density [7]. The construction of canals and dams has further fragmented rivers, reducing water availability downstream [8]. Climate change is anticipated to worsen the situation through changing rainfall patterns and increasing droughts [9].

The human response to this challenge has shifted from the more traditional approach of rainwater harvesting to huge projects like dams, reservoirs, recharging schemes, and desalination plants. Nevertheless, these projects have negative impacts. Dams create a cumulative geomorphological impact in a whole river system [10], and inadequate diversion can disrupt the natural hydrological process and dry up rivers and streams [11]. Even though the construction of reservoirs provides significant storage capacity for the provision of water, irrigation, and electricity, these systems create changes to the flow, sedimentation, and aquatic life. The effective running of reservoirs demands a delicate balance between the advantages of storage and flood control.

Despite considerable progress in optimisation-based and simulation-based reservoir operation, which is reviewed in Section 2, three gaps motivate the present work. First, many published models rely on optimisation algorithms or single neural-network architectures that require substantial calibration data and are seldom translated into a directly usable, hour-by-hour operating rule for spillway gates. Second, conventional field practice at many Indian reservoirs, including Shetrunji, still depends on manual reference tables for the level-storage and gauge-discharge relationships, which introduce interpolation error and are slow to apply during high-inflow events. Third, an existing study of the Shetrunji Reservoir predicted water level from meteorological inputs using artificial neural networks [12] but neither automated the departmental lookup workflow nor coupled a modern gradient-boosting predictor to an explicit, safety-aware gate-operation schedule. The specific research gap addressed here is therefore the absence of an integrated, data-driven framework that replaces manual lookup with validated regression models, forecasts short-horizon water level with an ensemble learner, and converts those forecasts into a tiered gate-operation rule for a real, gate-controlled reservoir in an arid Indian setting.

To address this gap, this study proposes and compares three techniques using actual data

from the Shetrunji Reservoir, using different statistical criteria to evaluate these techniques. The objectives of the present study are to develop reservoir water level forecasting models within operational periods; to automate the conventional lookup technique for water level and gate operation management decisions; to use machine learning for inflow-based water level prediction; to compare conventional approach and machine learning approach quantitatively; and to propose gate operation policies for flood risk management.

2. Related Works

Research on reservoir operation has drawn on optimisation theory, hydrological simulation, and, increasingly, machine learning. Meema et al. [13] optimised the real-time operation of a large-scale reservoir in Thailand by coupling adaptive inflow prediction with medium-range ensemble precipitation forecasts, showing that forecast-informed operation improves flood control. Yu et al. [14] addressed real-time flood control under forecasting uncertainty through multi-objective robust optimisation, and comparable machine-learning studies have combined back-propagation neural networks, long short-term memory networks, and extreme learning machines within stacking ensembles to raise short-term inflow-forecast accuracy. Rezay Nazarzadeh et al. [15] examined optimal pre-release strategies for real-time flood control at the Dez Reservoir while explicitly accounting for forecast uncertainty.

Evolutionary and heuristic optimisation has also featured prominently. Chang et al. [16] formulated a constrained genetic algorithm for multi-use operation of the Shih-Men Reservoir that respected ecological base-flow requirements, and Malekmohammadi et al. [17] integrated genetic algorithms with hydraulic flood routing and a K-Nearest Neighbour scheme for real-time flood management in river-reservoir systems. Russell and Campbell [18] explored fuzzy programming for deriving operating rules, while Kumar and Yadav [19] benchmarked Teaching-Learning-Based Optimisation and the Jaya algorithm on reservoir-operation problems. Al-Nouti et al. [20] synthesised this literature in a comprehensive review of machine-learning models for reservoir operation, cataloguing neural networks, genetic algorithms, decision trees, and ensemble methods, and identifying persistent gaps in validation and operational deployment.

In India, Thomas et al. [21] carried out the analysis of the impacts of climate change on Narmada basin through SWAT-based modeling and proved that the integrated

reservoir operation may mitigate irrigation and hydropower deficiencies, thus emphasizing the importance of adaptation measures. Anandharuban et al. [22] used box-modeling technique with HEC-HMS for the operational management of the reservoir in case of extreme precipitation events in Chennai region, and Ahmed and Sarma [23] designed the genetic algorithm-based operating policy for multipurpose reservoir on Pagladia river. The most relevant to this study are the research works by Bhatt et al. [12] who developed artificial neural networks to forecast the water level in Shetrunji River Dam based on rainfall, temperature, and evaporation. These works prove the efficiency of data-driven approaches but do not consider incorporating a validated rating curve model, a modern gradient-boosting forecaster, and an explicit gate operation policy for Shetrunji Reservoir, which is what we pursue in this study.

3. Proposed Work

This section describes the study area, the data used, the overall methodological workflow, and the three modelling approaches developed in this work, with all variables and parameters defined.

3.1 Study Area

The Shetrunji Reservoir, located near Palitana in Bhavnagar district, Gujarat, India, was selected because it has overflowed in five consecutive years, which makes its gate operation operationally critical. The Shetrunji is a principal river of the Saurashtra region, rising in the Chachai hills of Junagadh district at about 380 m above mean sea level and flowing eastward for 182 km before discharging into the Gulf of Cambay. It drains an area of 5514 square kilometres between longitudes 70 degrees 50 minutes and 72 degrees 10 minutes east and latitudes 21 degrees 00 minutes and 21 degrees 47 minutes north [24]. Nine tributaries longer than 15 km join the river, namely Safara, Shel, Kharai, and Talaji on the right bank and Satali, Thebu, Gagadia, Rajawal, and Kharo on the left bank [24]. Commissioned in 1959, the reservoir supplies irrigation and municipal water and has a total storage capacity of 415 million cubic metres, a spread area of 67.34 square kilometres at full reservoir level, and a catchment of 4317 square kilometres [25]. Figure 1 shows the location map of the Shetrunji River basin in Gujarat, India and a photograph of the dam with its spillway gates [25].

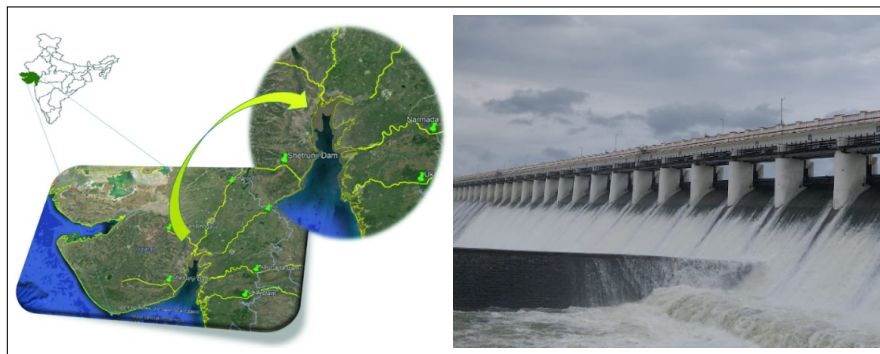


Figure 1. Shetrunji dam

Table 1. Shetrunji reservoir project, dam, spillway, and reservoir specifications [25]

Parameter	Value	Parameter	Value
Location	Palitana, Bhavnagar	Dam type	Earthen and masonry
Purpose	Irrigation	Foundation	Black trap
River	Shetrunji	Maximum height	32.02 m
Catchment area	4317 km ²	Top length	3895 m
Mean annual runoff	361.79 Mm ³	Concrete volume	0.0115 Mm ³
Mean annual rainfall	552 mm	Masonry volume	0.2917 Mm ³
Construction period	1955–1965	Earthwork volume	0.3329 Mm ³
Spillway type	Ogee shaped	Area at FRL	67.34 km ²
Spillway length	646 m	Gross storage	415.41 Mm ³
Energy dissipater	Roller bucket	Effective storage	374 Mm ³
Maximum discharge	7080 m ³ /s	Villages submerged	11 full, 13 partial
Gates	59 vertical, 8.84 × 0.91 m	Full reservoir level	1.53

3.2 Data Description

Hydrological records were obtained from Irrigation Sub-Division No. 01, Palitana, under the Narmada Water Resources, Water Supply and Kalpsar Department, Government

of Gujarat. The material includes hand-written registers kept by site engineers as well as computerized records generated for calculations. The variables include: Reservoir level in meters, Inflow height in feet, Discharge in cusec, Gross storage in million cubic feet, and 24 hours total rainfall in mm. The time interval is hourly when it rains and daily when there is no rain.

The dataset employed for this research was from the five hydrological years from 2020 to 2024, which is the period during which the reservoir experienced several instances of spill-over. Post-digitization and clean-up, the level-storage rating curve used in developing the Quadratic Regression model included the pairs of data for reservoir water level and gross storage from the department, while the gauge-discharge rating curve used in developing the K-Nearest Neighbours model included the pairs of data for the gauging and discharge of the reservoir. The dataset for water level forecast using the XGBoost algorithm included the daily and event-based values of reservoir water level and inflow, among others. In each case, the observations were randomly divided into 80 percent for training and 20 percent for testing using the same random seed number, thereby guaranteeing that the values of the coefficient of determination, the mean square error, and the root mean square error cited below refer to the test subset rather than the training one. Preprocessing steps included structuring of manually recorded data into spreadsheets, unit conversion from feet and inches into decimal feet, imputation of missing values, and creation of two lookup tables.

3.3 Methodological Workflow

The overall workflow (Figure 2) proceeds in five stages. In the first stage, data acquisition, the manual registers and departmental rating tables are collected from the Palitana sub-division. In the second stage, preprocessing, the entries are digitized, units are standardised, missing values are treated, and the level-storage and gauge-discharge rating tables are assembled. In the third stage, model development, three parallel approaches are built, namely automation of the existing lookup logic, the Intermediate Machine Learning scheme combining Quadratic Regression for the level-storage relationship with K-Nearest Neighbours regression for the gauge-discharge relationship, and the XGBoost water-level forecaster, each trained on the 80 percent training partition. In the fourth stage, evaluation, every model is scored on the 20 percent test partition and the three approaches are compared. In the fifth

stage, application, the validated forecasts are passed to a tiered gate-operation rule that issues gate-opening recommendations. The arrows in Figure 2 indicate the flow of data and decisions between these stages.

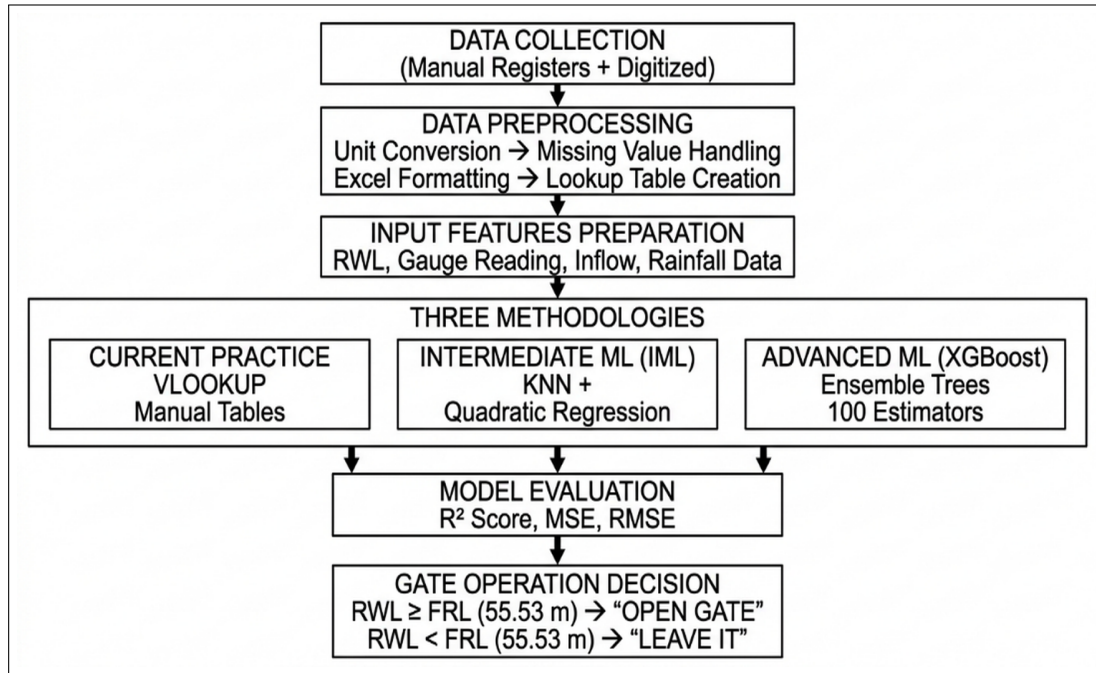


Figure 2. Flow diagram of the methodological workflow

3.4 Current Practice Automation

The conventional operating logic was codified into an algorithm using the programming language Python. If $R(t)$ is the height of the water in the reservoir in metres at time t , and $I(t)$ is the inflow discharge in cusec, then $S(R)$ is the gross storage function. Given the current level R_0 , the initial gross storage S_0 is retrieved from the lookup table as $S_0 = S(R_0)$. The volume added over one hour is then computed as shown in Equation (1).

$$V = I \times 3600 \times 10^{-6} \quad (1)$$

In Equation (1), V is the added volume in million cubic feet, the factor 3600 converts the hourly inflow rate to a volume, and the factor 10^{-6} converts the result to million cubic feet. The gross storage is updated as shown in Equation (2).

$$S_1 = S_0 + V \quad (2)$$

The new level R_1 is then determined by inverse lookup such that $S(R_1) = S_1$, and the

baseline gate-operation decision is applied as shown in Equation (3).

$$Gate\ Status = OPEN\ if\ R_1 \geq R_{FRL},\ CLOSED\ if\ R_1 < R_{FRL} \quad (3)$$

In Equation (3), R_{FRL} is the full reservoir level, which equals 55.53 m for the Shetrunji Reservoir. This baseline rule is deliberately simple, and a refined, tiered rule is developed in Section 4.4.

3.5 Level-Storage Relationship Using Quadratic Regression

The storage-elevation relationship is non-linear. Let X denote the reservoir water level in metres and let Y denote the gross storage in million cubic feet. The quadratic regression model is given in Equation (4).

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon \quad (4)$$

In Equation (4), β_0 is the intercept, β_1 and β_2 are the linear and quadratic coefficients respectively, and ε is the random error term. The coefficients are estimated by ordinary least squares, that is, by minimising the residual sum of squares over the n training observations as shown in Equation (5).

$$\min \sum_i (Y_i - \beta_0 - \beta_1 X_i - \beta_2 X_i^2)^2, \quad i = 1 \dots n \quad (5)$$

In Equation (5), X_i and Y_i are the i -th observed reservoir water level and gross storage values respectively. Model quality was assessed on the test partition using the coefficient of determination defined in Equation (6).

$$R^2 = 1 - [\sum_i (Y_i - \hat{Y}_i)^2] / [\sum_i (Y_i - \bar{Y})^2] \quad (6)$$

In Equation (6), $\hat{Y}_i$ is the model prediction for observation i and $\bar{Y}$ is the mean of the observed values of Y .

3.6 Gauge-Discharge Relationship Using K-Nearest Neighbours Regression

The gauge-discharge relationship was modelled with K-Nearest Neighbours regression. For a gauge reading x , the predicted discharge $\hat{y}$ is the weighted mean of the discharges of its k nearest neighbours, as shown in Equation (7).

$$\hat{y} = (1/k) \sum y_i, \quad i \in N_k(x) \quad (7)$$

In Equation (7), $N_k(x)$ is the set of the k training points closest to x and y_i denotes their observed discharges. Proximity is measured by Euclidean distance, which for this one-dimensional feature reduces to the expression in Equation (8).

$$d(x, x_i) = |x - x_i| \quad (8)$$

Hyperparameters were tuned by grid search with five-fold cross-validation, selecting the parameter set θ that minimises the mean cross-validated error as shown in Equation (9).

$$\theta^* = \arg \min (1/5) \sum_j MSE(\theta, D_j), j = 1 \dots 5 \quad (9)$$

In Equation (9), D_j is the j -th validation fold. The search over values of k in the set $\{2, 3, 5, 7, 10\}$ and over uniform and distance weighting selected k equal to 5 with distance weighting.

3.7 Water-Level Forecasting Using XGBoost Regression

XGBoost, or Extreme Gradient Boosting, is an ensemble method that builds an additive sequence of M regression trees. Using reservoir water level and inflow as the input features x_i , the prediction for observation i is the sum of the tree outputs as shown in Equation (10).

$$\hat{y}_i = \sum f_m(x_i), m = 1 \dots M, f_m \in F \quad (10)$$

In Equation (10), F is the space of regression trees and f_i is the m -th tree. Training minimises the regularised objective given in Equation (11).

$$L(\varphi) = \sum_i l(y_i, \hat{y}_i) + \sum_m \Omega(f_m) \quad (11)$$

In Equation (11), l is the differentiable loss between the observed value y_i and the predicted value $\hat{y}_i$, and Ω penalises tree complexity as defined in Equation (12).

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2 \quad (12)$$

In Equation (12), T is the number of leaves in tree f , ω is the vector of leaf weights, and γ and λ are regularisation parameters that control the penalty per leaf and the L2 penalty on the leaf weights respectively. At iteration t , the objective is approximated by a second-order Taylor expansion as shown in Equation (13).

$$L^{(t)} \approx \sum_i [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (13)$$

In Equation (13), g_i and h_i are the first-order and second-order gradients of the loss with respect to the previous prediction. The forecast accuracy was quantified using the root mean squared error and the mean squared error, which are defined in Equation (14).

$$RMSE = \sqrt{[(1/n) \sum_i (y_i - \hat{y}_i)^2]}, \quad MSE = (1/n) \sum_i (y_i - \hat{y}_i)^2 \quad (14)$$

4. Results and Discussion

All three approaches were implemented in Python using the scikit-learn and XGBoost libraries and were evaluated on the held-out test partition, which comprises 20 percent of the observations, using the coefficient of determination, the mean squared error, and the root mean squared error.

4.1 Model Performance Evaluation

Though the manual look-up method works, there is a problem of approximation. In case the water level of the intermediate reservoir falls between two tabulated values, the engineer chooses the higher one, thereby overestimating the storage capacity. Automation of this process via Python did not help to get rid of the limitation of a look-up table that lies in its inherent discreteness and prompted the development of regression models described further. Table 2 below shows the performance of the two Intermediate Machine Learning models on the test set.

Table 2. Performance of the intermediate machine learning models on the test partition

Model	Equation or algorithm	R ²	MSE
Quadratic Regression	$Y = \beta_0 + \beta_1 X + \beta_2 X^2$	0.9999	1403.1950
KNN Regression	$\hat{Y} = \left(\frac{1}{k}\right) \sum_i \in N_k(x) y_i$	0.9995	789709.3244

The quadratic model explains 99.99 percent of the variance in gross storage as a function of reservoir water level, and the K-Nearest Neighbours model reproduces discharge from the gauge reading with a coefficient of determination of 0.9995. These very high values reflect the fact that both relationships are physical rating curves. The level-storage curve is fixed by the reservoir geometry and the gauge-discharge curve is fixed by the spillway rating, so both are smooth and near-deterministic, and the models are effectively learning a well-behaved calibration curve rather than a noisy stochastic process. The large absolute mean squared error

for the K-Nearest Neighbours model simply reflects the units of discharge and is small relative to the range of the discharge values.

Trained on reservoir water level and inflow, the XGBoost model attained a coefficient of determination of 0.9995, a root mean squared error of 0.0419 m, and a mean squared error of 0.00175 on the test partition. A root mean squared error below 5 cm is accurate for operational water-level forecasting near the full reservoir level threshold. As with the rating-curve models, this high accuracy should be read in context, because the one-hour-ahead water level is strongly determined by the current level plus a small inflow increment, so the predictive task is inherently well conditioned. Table 3 compares the three approaches.

Table 3. Comparative performance of all three approaches on the test partition

Approach	R ²	MSE	RMSE (m)	Key advantage
Current practice	Not applicable	Not applicable	Not applicable	No training required
Quadratic Regression	0.9999	1403.1950	Not applicable	Fits non-linear level-storage curve
KNN Regression	0.9995	789709.3244	Not applicable	Captures gauge-discharge rating
XGBoost	0.9995	0.00175	0.0419	Direct level forecast using an ensemble

Two points regarding validation follow from Table 3. First, because the reported metrics come from a held-out test set that was excluded from training, they are out-of-sample rather than in-sample scores. Second, the near-unity values of the coefficient of determination arise chiefly from the smooth, low-noise nature of the underlying rating relationships and from the short one-hour forecast horizon, and they should not be interpreted as evidence that the models would generalise unchanged to longer horizons, to other reservoirs, or to conditions outside the range observed between 2020 and 2024. Stronger independent validation, for example k-fold cross-validation across multiple hydrological years, testing on an entirely separate reservoir, and residual analysis under extreme-inflow conditions, is identified as necessary future work in

Section 5. The comparison nonetheless shows that XGBoost gives the lowest root mean squared error and mean squared error for direct level forecasting, while the Intermediate Machine Learning models remain the most transparent choice for the individual rating relationships.

4.2 Rainfall and Inflow Relationship

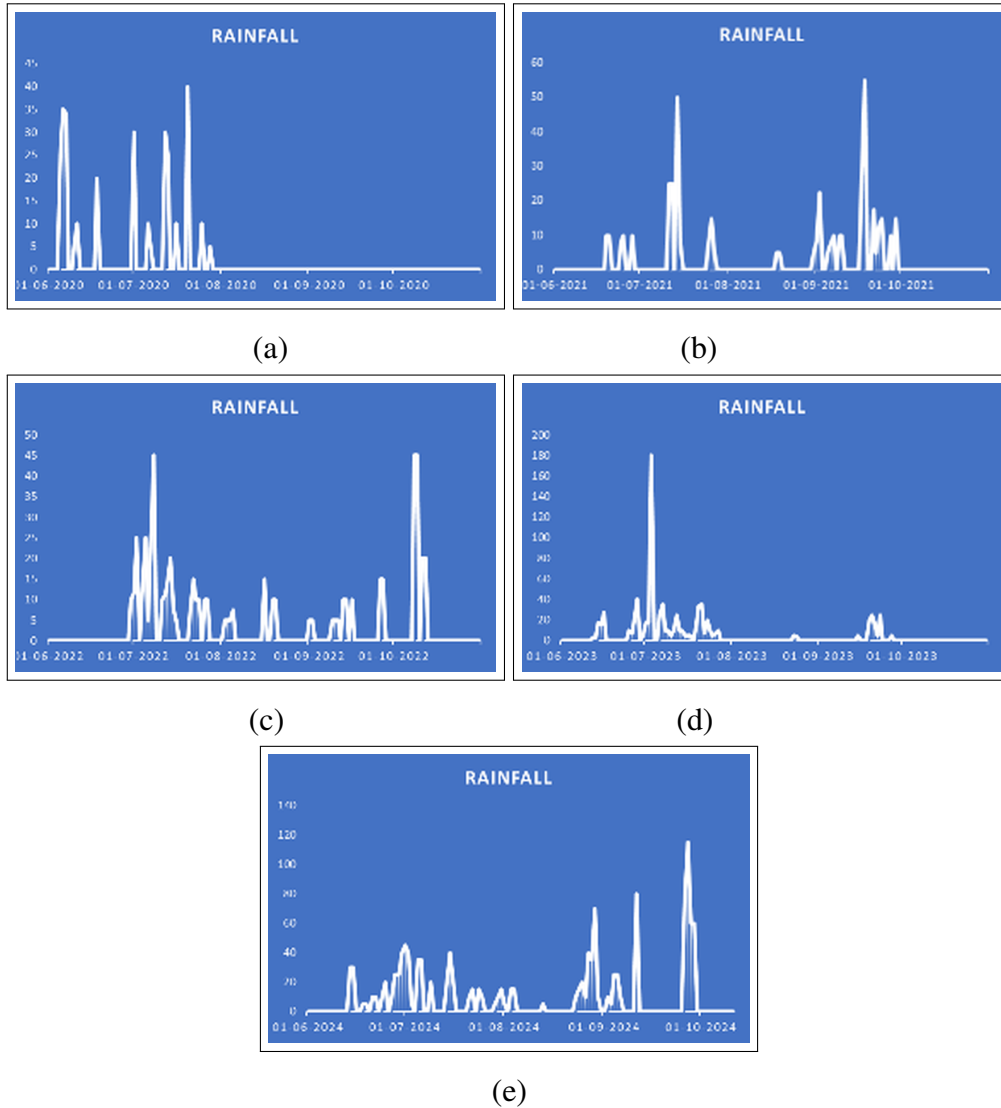


Figure 3. Monthly rainfall and reservoir water level comparison from 2020 to 2024. (a) 2020, (b) 2021, (c) 2022, (d) 2023, (e) 2024

Precipitation and reservoir water level data for the period 2020 to 2024 show clear associations between upstream rainfall and reservoir response. Figure 3 presents the year-by-year monthly comparison of rainfall and reservoir water level, with one panel for each year of the study period.

The analysis of Figure 3 indicates that rainfall strongly influences the reservoir water

level, but that the response is not governed by on-site precipitation alone. Upstream rainfall over the catchment of 4317 square kilometres contributes substantially to the inflow, and four to five smaller upstream dams regulate the release into the main reservoir, which produces cascading effects. In addition, the hilly terrain promotes rapid runoff, which further complicates reservoir management.

4.3 Prediction Results

The XGBoost model was applied to predict the reservoir water level for 2024 using the inflow and the current reservoir water level as inputs. Table 4 lists representative predictions drawn from the test partition.

Table 4. Sample reservoir water level predictions obtained from the XGBoost model for 2024

Actual RWL (m)	Inflow (ft)	Reference RWL (m)	Predicted RWL (m)	Error (m)
49.75	0	49.80	49.800110	0.000110
49.70	0	49.75	49.750183	0.000183
49.60	0	49.70	49.700451	0.000451
49.55	1	49.60	49.602345	0.002345
49.75	2	49.80	49.800156	0.000156
49.95	1	49.95	49.949799	0.000201
50.15	3.25	50.20	50.000435	0.000435
52.05	2	52.20	52.200077	0.000077
52.85	3.5	52.85	52.851234	0.001234
54.90	14	54.95	54.950116	0.000116

The maximum error in this sample is below 0.003 m, which is consistent with the aggregate root mean squared error of 0.0419 m obtained over the full test partition.

4.4 Gate Operation Recommendations

The basic decision rule expressed in Equation (3) opens the gates as soon as the predicted water level attains the water level of the reservoir. A one-threshold approach is not, however, sufficient to determine how many out of 59 gates to open and in what sequence as well as whether to escalate since it just indicates that gates need to be opened. Transitioning from predicting to optimising the operation of the gates entails that the binary decision rule be extended to the sequential rule depicted in Table 5. The sequential rule stratifies the water level, sequences gates from the middle towards the ends since the gates in the middle are easier to operate than the end gates and increases the number of gates depending on the water level. The forecasted water level is available approximately an hour before the event occurs.

Table 5. Tiered gate-operation schedule based on the predicted reservoir water level

Scenario	Water level condition	Action	Gate priority
Normal	$RWL < 55.53 \text{ m}$	Maintain current status	None
Warning	$55.53 \text{ m} \leq RWL < 55.80 \text{ m}$	Open central gates	Central gates 1 to 10
Critical	$55.80 \text{ m} \leq RWL < 56.00 \text{ m}$	Open additional gates	Expand outward
Emergency	$RWL \geq 56.00 \text{ m}$	Open all gates	All 59 gates

This tiered rule maintains the clarity associated with the level threshold of the entire reservoir while incorporating sequencing, escalation, and safety considerations that would make it operational. The strengths associated with this rule include the removal of human error, consistency and repeatability of the process, priority given to central gates to reduce stress on the structure, and an early warning system of about one hour.

4.5 Discussion

In all three methods, the errors are below 5 cm. XGBoost model provides the best estimate of the water level directly with root mean squared error of 0.0419 m and coefficient of determination of 0.9995. Quadratic Regression method estimates the water-storage elevation

relation better with a coefficient of determination of 0.9999. The analysis of the data for the period 2020 to 2024 confirms that upstream rainfall over the catchment of 4317 square kilometres, modulated by four to five upstream dams, drives the inflow through cascading releases, which explains why on-site rainfall alone does not fully determine the reservoir water level. Compared with the earlier artificial neural network modelling reported for the Shetrunji Reservoir [12] and with classical time-series approaches, the gradient-boosting model handles the non-linear relationship between level and inflow well within the tested range. The main limitations of the present study are manual recording error, limited temporal resolution, validation on a single reservoir, unmodelled upstream-dam releases, and the three-inch least count of the inflow readings.

5. Conclusion

This study developed and compared three approaches for reservoir water-level prediction and gate-operation optimisation at the Shetrunji Reservoir using observed data for the period 2020 to 2024. These approaches were the automation of the current lookup practice, an Intermediate Machine Learning scheme combining Quadratic Regression for the level-storage relationship with K-Nearest Neighbours regression for the gauge-discharge relationship, and XGBoost regression for direct water-level forecasting. On the held-out test partition, the Quadratic Regression model reached a coefficient of determination of 0.9999 for the level-storage relationship, the K-Nearest Neighbours model reached 0.9995 for the gauge-discharge relationship, and the XGBoost model attained a root mean squared error of 0.0419 m, a coefficient of determination of 0.9995, and a mean squared error of 0.00175. In addition to providing predictions, the study also offers a hierarchy-based gate-operation rule which goes beyond the basic rule based on reservoir level through water-level intervals, center-to-outside order, and escalation, transforming the forecasts into action and decision-taking with safety considerations. Analysis of rainfall and inflow data indicates that the upstream rainfall across the catchment, through which water flows through a few upstream dams, has a major impact on the reservoir behavior. In summary, the findings suggest that the combination of validated data-driven models with the current hydrological process can enhance the precision and consistency of forecasts as well as risk mitigation for floods. Future work will focus on extending the forecast period, considering upstream dam

outflow, and optimizing the gate openings.

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